

Daily Sequences Drill #1 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Arithmetic Sequences & Series · n th-term & Sum Formulas · S_n as a Quadratic in n .

Section A Rapid Recognition

1. **Answer:** $3n + 2$

Method: $a = 5, d = 3$. $a_n = a + (n - 1)d = 5 + 3(n - 1) = 3n + 2$.

Investigate further: Check: $a_1 = 5$ ✓, $a_4 = 14$ ✓. Every AP's n th term is linear in n — the coefficient of n is always the common difference.

2. **Answer:** -50

Method: $a_{20} = 7 + 19(-3) = 7 - 57 = -50$.

Investigate further: A negative common difference just means the “ $n - 1$ steps” subtract instead of add. The formula never changes.

3. **Answer:** 265

Method: $S_{10} = \frac{10}{2}(2(4) + 9(5)) = 5(8 + 45) = 5(53) = 265$.

Investigate further: Cross-check with $S_n = \frac{n}{2}(a_1 + a_n)$: $a_{10} = 4 + 9(5) = 49$, $S_{10} = 5(4 + 49) = 265$. ✓

4. **Answer:** 3, 7, 11; arithmetic with $d = 4$

Method: $a_1 = 3, a_2 = 7, a_3 = 11$. Consecutive differences are both 4, so the sequence is arithmetic.

Investigate further: Any n th term of the form $pn + q$ is automatically arithmetic with $d = p$ — this is worth recognising instantly rather than re-deriving each time.

5. **Answer:** 4

Method: $a_5 - a_1 = 4d = 16 \Rightarrow d = 4$.

Investigate further: In general, $a_m - a_k = (m - k)d$ — the number of “steps” between any two terms is the index difference, not one of the indices themselves.

6. **Answer:** 1275

Method: $S_{50} = \frac{50 \cdot 51}{2} = 1275$.

Investigate further: This is Gauss's schoolroom trick: pair the 1st and 50th, 2nd and 49th, etc. — each pair sums to 51, and there are 25 pairs.

7. Answer: 4

Method: $a_1 = S_1 = 1^2 + 3(1) = 4$.

Investigate further: More generally $a_n = S_n - S_{n-1}$ for $n \geq 2$, but a_1 is just S_1 on its own — a common slip is applying the subtraction formula to $n = 1$ too, which needs S_0 (always 0) instead.

8. Answer: True

Method: $a_n = a_1 + (n - 1)d$ is a sum of integers whenever a_1, d, n are integers, so every term is an integer.

Investigate further: The converse is false: a sequence of all integers need not be arithmetic (e.g. $1, 2, 4, 8, \dots$).

9. Answer: 4

Method: 9th term – 3rd term is 6 steps of d : $37 - 13 = 24 = 6d \Rightarrow d = 4$.

Investigate further: Always count steps as (index difference), not (larger index). Between the 3rd and 9th term there are $9 - 3 = 6$ gaps, not 9 or 3.

10. Answer: 5

Method: 6th term – 1st term is 5 steps: $29 - 4 = 25 = 5d \Rightarrow d = 5$.

Investigate further: Same step-counting idea as A9, now anchored at the first term directly.

Section B Manipulation Drills**1. Answer: $d = 4$, $S_{15} = 465$**

Method: $l = a + (n - 1)d \Rightarrow 59 = 3 + 14d \Rightarrow d = 4$. $S_{15} = \frac{15}{2}(3 + 59) = \frac{15}{2}(62) = 465$.

Investigate further: Whenever you're given first term, last term, and count, $S_n = \frac{n}{2}(a + l)$ is faster than finding d first and building back up through $S_n = \frac{n}{2}(2a + (n - 1)d)$.

2. Answer: B) 70

Method: $S_8 - S_4$ is exactly the sum of terms 5 through 8, by definition of S_n as a running total: $100 - 30 = 70$. No need to ever find a or d .

Investigate further: The trap is trying to solve for a, d from the two sums first — possible, but a detour. $S_8 - S_4$ is a direct subtraction of two given numbers.

3. Answer: Proof by the AP sum formula.

Method: $3, 6, 9, \dots, 3n$ is an AP with $a = 3$, $d = 3$, last term $3n$, and n terms. $S_n = \frac{n}{2}(a + l) = \frac{n}{2}(3 + 3n) = \frac{3n(n+1)}{2}$. $\square$

Investigate further: This is exactly $3 \times (1 + 2 + \dots + n)$ — scaling every term of an AP by a constant k scales the sum by k too, since S_n is linear in the terms.

4. Answer: A) 15

Method: For an AP with an odd number of terms $n = 2k + 1$, the sum equals n times the middle term

(the terms pair up symmetrically around it, and the middle term is left over unpaired but equal to the pair-average). Here $315 = 21 \times (\text{middle term}) \Rightarrow \text{middle term} = 15$.

Investigate further: This only works because 21 is odd — for an even-length AP there's no single middle term, and you'd need $S_n = \frac{n}{2}(a_1 + a_n)$ instead, which needs both endpoints.

5. Answer: 1 (the 4th term)

Method: $a_n = 10 - 3(n - 1) > 0 \Rightarrow n < \frac{13}{3} \approx 4.33$, so the last positive term is $a_4 = 10 - 9 = 1$ (and $a_5 = -2 < 0$ confirms the sequence has already crossed zero).

Investigate further: Always check the boundary term itself, not just the inequality's cutoff — $n < 4.33$ tells you $n = 4$ is the last valid integer, but you still need to confirm a_4 is actually positive (it could in principle have landed exactly on 0).

6. Answer: D) All terms are positive.

Method: $d > 0$ guarantees the sequence is strictly increasing (A, true) and therefore unbounded above (C, true), and by definition $a_{n+1} - a_n = d$ always (B, true). But the first term a_1 can be any real number, including very negative — e.g. $a_1 = -100, d = 1$ has plenty of negative terms. So D is the one that need not hold.

Investigate further: $d > 0$ constrains the *shape* of the sequence, not its starting position.

7. Answer: A) 6

Method: Let the sequences be $a_n = a_1 + (n - 1)d$ and $b_n = b_1 + (n - 1)d$ (same d). Then $a_n - b_n = a_1 - b_1 = 6$ for every n — the $(n - 1)d$ terms cancel exactly because the common difference is shared.

Investigate further: If the common differences were different, the gap would grow (or shrink) linearly in n instead of staying constant — try it with $d_1 = 2, d_2 = 3$ to see the gap become $6 + (n - 1)$.

8. Answer: C) 5

Method: $a_3 + a_5 = (a + 2d) + (a + 4d) = 2a + 6d = 32 \Rightarrow a + 3d = 16$. Subtracting $a_3 = a + 2d = 11$: $(a + 3d) - (a + 2d) = 16 - 11 \Rightarrow d = 5$.

Investigate further: Once $d = 5$, back-substitute: $a = 11 - 2(5) = 1$. Always sanity-check by plugging both original conditions back in: $a_3 = 1 + 10 = 11 \checkmark$, $a_3 + a_5 = 11 + 21 = 32 \checkmark$.

9. Answer: D) (2, 4)

Method: Need $a + 4d = 17$ with a, d positive integers. Check each: $13 + 4(1) = 17 \checkmark$, $9 + 4(2) = 17 \checkmark$, $5 + 4(3) = 17 \checkmark$, but $2 + 4(4) = 18 \neq 17$ — D fails the defining equation outright.

Investigate further: The three valid pairs follow the pattern “ a drops by 4 as d rises by 1” (13, 9, 5, 1, ...) — option D's $a = 2$ breaks that pattern, which is worth noticing as a sanity check even before computing.

10. Answer: B) $n = 3$

Method: AP: 2, 5, 8, 11, 14, 17. GP: 2, 4, 8, 16, 32, 64. Comparing term by term, both give 8 at the 3rd term.

Investigate further: After $n = 3$ the GP overtakes the AP permanently and they never meet again — geometric growth eventually dominates linear growth no matter the starting ratio, a theme this pillar returns to when comparing arithmetic and geometric series directly.

Section C Substitution & Structure

1. **Answer:** A) none of them

Method: Writing $S_n = \frac{n}{2}(2a + (n-1)d) = 15$ gives $n(2a + (n-1)d) = 30$, so n must divide 30. Counterexamples kill every statement at once: with $n = 3, a = 4, d = 1$ the sequence is 4, 5, 6 (sum 15) — $n = 3$ is odd (kills I) and $a = 4$ is even (kills II). With $n = 3, a = 3, d = 2$ the sequence is 3, 5, 7 (sum 15) — $d = 2$ is even (kills III). No single parity pattern is forced.

Investigate further: This is the real trap of “must be true” questions: checking one valid example never proves a statement false, but finding *one* counterexample per statement is enough to rule each one out individually.

2. **Answer:** A) arithmetic with common difference 5

Method: $c_n = (2n + 1) + (3n - 2) = 5n - 1$, which is linear in n with slope 5 — so (c_n) is arithmetic with $d = 5$.

Investigate further: This generalises: the sum of two arithmetic sequences is always arithmetic, with common difference equal to the sum of the two original common differences ($2 + 3 = 5$ here). The same is not true for products, which is exactly what C3 tests next.

3. **Answer:** C) arithmetic only if $d = 0$

Method: $a_n = a + (n-1)d$, so $b_n = (a + (n-1)d)^2$ is a quadratic in n with leading coefficient d^2 . A quadratic sequence is arithmetic (i.e. linear in n) only if its leading coefficient vanishes: $d^2 = 0 \Rightarrow d = 0$.

Investigate further: This is the flip side of C2: sums of AP's stay linear in n , but squaring introduces an n^2 term unless the original sequence was already constant.

4. **Answer:** A) d must be positive

Method: $S_6 = 6a + 15d$. Sum of terms 7–10 is $4a + (6 + 7 + 8 + 9)d = 4a + 30d$. Setting equal: $6a + 15d = 4a + 30d \Rightarrow 2a = 15d \Rightarrow a = 7.5d$. For $a > 0$ (all terms positive requires at least $a_1 > 0$), d must be positive too, since $a = 7.5d$ forces d and a to share a sign.

Investigate further: Positivity of *all* terms is a stronger condition than just $a > 0$ — with $d > 0$ every later term is even bigger, so once $a_1 > 0$ the rest follow automatically.

5. **Answer:** A) (5, 19)

Method: Need $d \mid (100 - a)$. Check: $100 - 5 = 95 = 19 \times 5 \checkmark$; $100 - 6 = 94$, $94/17$ is not an integer; $100 - 7 = 93$, $93/23$ is not an integer; $100 - 8 = 92$, $92/13$ is not an integer. Only A works.

Investigate further: This is the same divisibility idea as “does an AP hit a target value” in general: $100 = a + (n-1)d$ for some positive integer n exactly when $d \mid (100 - a)$.

6. **Answer:** A) d must equal 0

Method: $S_{2n} = n(2a + (2n-1)d)$ and $2S_n = n(2a + (n-1)d)$. Setting them equal and dividing by

$n > 0$: $2a + (2n - 1)d = 2a + (n - 1)d \Rightarrow nd = 0$. Since $n > 0$, this forces $d = 0$.

Investigate further: This is a genuinely surprising result: doubling the term count can only double the sum if the sequence isn't actually changing at all. It's the reason S_n growing quadratically (C7) matters — a truly arithmetic (non-constant) sequence's sum grows faster than linearly, so doubling n more than doubles S_n .

7. Answer: C) quadratic in n

Method: $S_n = \frac{n}{2}(2a + (n - 1)d) = \frac{d}{2}n^2 + (a - \frac{d}{2})n$. Since $d \neq 0$, the n^2 coefficient is nonzero, so S_n is genuinely quadratic (not linear, and it does have a linear term unless $a = \frac{d}{2}$).

Investigate further: This is the single most useful fact in the whole sheet: any S_n you're handed that's a quadratic with no constant term secretly comes from an AP, and you can read off d and a directly from its coefficients — exactly what C8 and D4 ask you to do.

8. Answer: B) 6

Method: Matching $S_n = 3n^2 - n$ to $S_n = \frac{d}{2}n^2 + (a - \frac{d}{2})n$: $\frac{d}{2} = 3 \Rightarrow d = 6$, and $a - \frac{d}{2} = -1 \Rightarrow a = 2$.

Investigate further: The most common slip is answering 3 (forgetting the $\frac{1}{2}$ in front of d) or -1 (reporting the coefficient of n instead of solving for d). Always solve $\frac{d}{2} = (\text{coefficient of } n^2)$, not $d = (\text{coefficient of } n^2)$.

Section D Challenge

1. Answer: A) $a = -8d$

Method: $S_6 = S_{11} \Rightarrow \frac{6}{2}(2a + 5d) = \frac{11}{2}(2a + 10d) \Rightarrow 3(2a + 5d) = 5.5(2a + 10d) \Rightarrow 6a + 15d = 11a + 55d \Rightarrow -5a = 40d \Rightarrow a = -8d$.

Investigate further: The real TMUA question (2018 Paper 1 Q2) uses $S_5 = S_8$ and lands on $a = -\frac{38}{3}d$ — same technique, uglier numbers. Try that version by hand to see how the method scales.

2. Answer: C) $n = 15$

Method: The sequence $2, 5, \dots, 80$ has 27 terms, $a_i = 3i - 1$. Two terms with indices i, j sum to 82 exactly when $i + j = 28$. Pairing indices 1–27 by this rule gives 13 genuine pairs $(1, 27), (2, 26), \dots, (13, 15)$, plus one unpaired index (14, whose partner would be itself). A selection can safely take one element from each of the 13 pairs plus the unpaired element — 14 terms with no pair summing to 82. Any 15th term forces a complete pair. So the smallest guaranteeing n is 15.

Investigate further: This is the Pigeonhole Principle in disguise: the “pairs” are the pigeonholes, and once you must place more elements than there are safe one-per-pair slots (plus the singleton), a full pair is forced. The real TMUA question (2022 Paper 2 Q8) uses the AP $1, 4, \dots, 70$ and target sum 74, landing on $n = 14$ — work through why the singleton index shifts between the two versions.

3. Answer: d must be a multiple of 6.

Method: *Mod 2:* since $p > 3$ is prime, p is odd. If d were odd, $p + d$ would be even and greater than 2, hence composite — contradiction. So d is even.

Mod 3: since $p > 3$, $p \not\equiv 0 \pmod{3}$. If $d \not\equiv 0 \pmod{3}$, then d is coprime to 3, so $p, p + d, p + 2d$ take all three residues mod 3 (adding a unit repeatedly cycles through every residue) — meaning one of them is $\equiv 0 \pmod{3}$. That term is greater than 3 and divisible by 3, hence composite — contradiction. So $3 \mid d$.

Combining: $2 \mid d$ and $3 \mid d \Rightarrow 6 \mid d$. $\square$

Investigate further: The smallest example is $p = 5, d = 6$: 5, 11, 17, 23 — all prime. Try to find the next one by hand, and notice how quickly $d = 6$ stops being sufficient on its own (having $6 \mid d$ is necessary, not sufficient, for four primes in AP).

4. **Answer:** A) -4

Method: $a_n = S_n - S_{n-1} = A(2n-1) + B$ (expand and simplify $An^2 + Bn - A(n-1)^2 - B(n-1)$). With $A = 5$: $a_{10} = 5(19) + B = 95 + B = 91 \Rightarrow B = -4$.

Investigate further: This formula $a_n = A(2n-1) + B$ is worth memorising directly: it's the general rule for reading off any term of an AP straight from its $S_n = An^2 + Bn$ form, without ever finding S_{n-1} by hand each time.

5. **Answer:** B) a_n is a quadratic function of n .

Method: If the first differences $a_{n+1} - a_n$ form an AP, then the *second* differences of (a_n) are constant — and a sequence with constant second differences is exactly a quadratic function of n (the discrete analogue of a constant second derivative meaning a parabola).

Investigate further: This is C7 in reverse: there, an AP's partial sum S_n turned out to be quadratic in n because S_n 's first differences ($= a_n$) form an AP. This question generalises that observation to any sequence, not just partial sums — the same idea will resurface when this pillar looks at general recurrence relations.

Top 5 Patterns — Worksheet #1

- 1. nth-term formula:** $a_n = a + (n-1)d$. The foundation everything else builds on — always count *steps* (index differences), never raw indices. (see A1, A2, A5, A9, A10)
- 2. Sum formula:** $S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + l)$. Use the $(a + l)$ form whenever you're given the first and last term directly — it skips finding d entirely. (see A3, A6, B1)
- 3. S_n is quadratic in n , with no constant term.** $S_n = \frac{d}{2}n^2 + (a - \frac{d}{2})n$ — read d and a straight off the coefficients, and extract any term via $a_n = A(2n-1) + B$. (see C7, C8, D4, D5)
- 4. Middle-term trick for odd-length AP's.** When n is odd, $S_n = n \times (\text{middle term})$ — no need to find a or d at all. (see B4)
- 5. Pairing/Pigeonhole within an AP.** To force two terms to sum to a target, pair up indices by that target and count the leftover singleton — the answer is always (safe pairs) + (singletons) + 1. (see D2)

Common Traps — Worksheet #1

- 1. Forgetting the $\frac{1}{2}$ in the sum formula.** Matching $S_n = 3n^2 - n$ to $\frac{d}{2}n^2 + \dots$ gives $d = 6$, not $d = 3$ — always solve for d from the *halved* coefficient. (see C8)
- 2. Trusting one example for a “must be true” claim.** A single valid (a, d, n) never proves a statement true — you need a counterexample to rule one out, or a full argument to confirm one always holds. (see C1)
- 3. Assuming every quadratic S_n -like formula is automatically a valid AP sum.** The

nth-term extraction $a_n = S_n - S_{n-1}$ only makes sense when $S_0 = 0$ is consistent with the formula — check there's no stray constant term. *(see D4)*

4. **Sign errors when solving simultaneous AP equations.** A negative common difference is completely valid — don't discard a solution just because $d < 0$ looks unusual. *(see D1, B6)*
5. **Declaring a sequence arithmetic from a few terms without proving it for general n .** Checking three consecutive differences match is suggestive, not a proof — squaring an AP (C3) is exactly the case where a plausible-looking pattern breaks down algebraically. *(see C3)*

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths